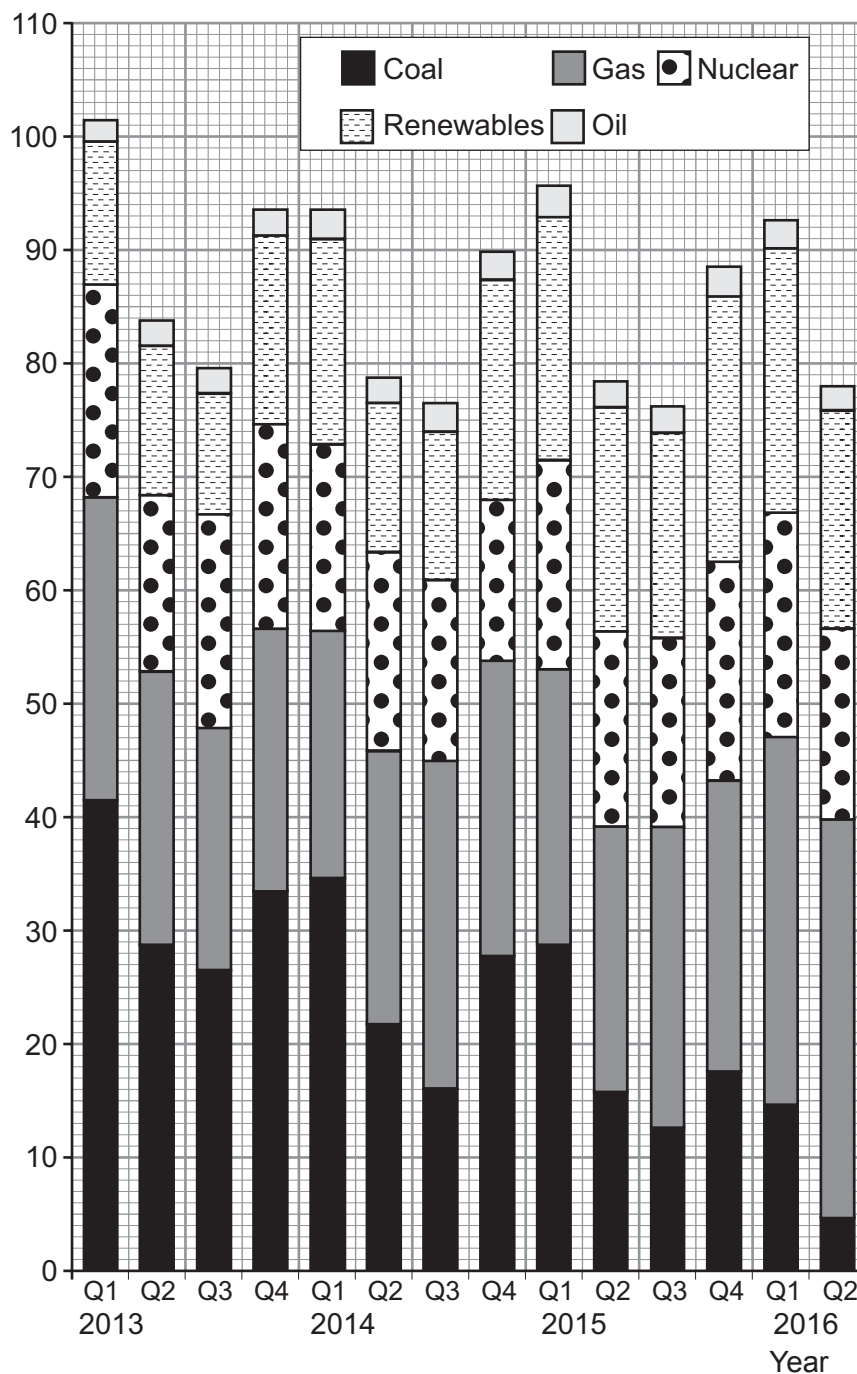


6. The Climate Change Act established a target for the UK to reduce its 1990 CO₂ emissions by at least 80 % by 2050. This target represents an appropriate contribution from the UK towards the agreed reduction in CO₂ emissions globally. It aims to limit the global temperature rise to as little as possible above 2 °C.

To ensure that regular progress is made towards this long-term target, the Act also established a system of five-yearly carbon budgets, to serve as stepping stones on the way.

The diagram below shows the sources of fuel used in the production of electrical energy between the first quarter of 2013 and the middle of 2016.

Generation (TWh)



KEY:

Q1 is 1st quarter of the year (Jan – Mar)

Q2 is 2nd quarter of the year (Apr – June)

Q3 is 3rd quarter of the year (July – Sept)

Q4 is 4th quarter of the year (Oct – Dec)



- (a) (i) Use the information in the diagram to describe trends in the use of individual fossil fuels and the total use of fossil fuels over the time shown. [3]

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- (ii) The use of nuclear fuel for the production of electricity between Q1 2013 and Q1 2016 remained roughly constant. Suggest possible reasons for this. [2]

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- (iii) Production of electricity from renewable sources remained roughly constant for Q2 2015 and Q2 2016. However, there was a reduction in the contribution by wind and a change in the contribution from solar power. Describe how the weather conditions were different in Q2 2016 compared to Q2 2015. [2]

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Examiner
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- (b) The first four carbon budgets, leading to 2027, have been set in law. The UK is currently in the third carbon budget period (2018-22). Meeting the fourth carbon budget (2023-27) will require that emissions be reduced by 50 % on 1990 levels by 2027.

The 1990 levels of CO₂ emissions were 3900 million tonnes.

Budget	CO ₂ emission targets from all sources	% reduction below 1990 levels for all sources
1st 5 yr period achievement (2008-12)	Down to 3018 million tonnes of CO ₂ by the end of 2012	
2nd 5 yr period target (2013-17)	Down to 2782 million tonnes of CO ₂ by the end of 2017	
3rd carbon target (2018-22)	Down to 2544 million tonnes of CO ₂ by the end of 2022	
4th carbon target (2023-27)	Down to 1950 million tonnes of CO ₂ by the end of 2027	50 % by the end of 2027

- (i) Calculate the targeted percentage drop in CO₂ emissions **from all sources** between 1990 and the end of 2022. Show your workings clearly. [2]

Percentage drop =

- (ii) Between 1990 and the end of 2022, the mean percentage drop in the use of **fossil fuels in the production of electricity** is expected to be 40 %. Comment on the progress being made in achieving the government’s overall target for CO₂ emissions by 2027. [2]

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END OF PAPER

